



Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8524**

AV:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	401	Seat No:	2022600046
Course Section:	1				



Name of the candidate. Pushap Raina

Candidate's UID number:

2	0	2	2	6	0	0	0	4	6
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Examination class room number:

4	0	1
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Invigilator's signature with date:

<u>8/10/25</u>

(To be filled in by the candidate)

EXAMINATION: BE MSE 2025 Tech
(For example FY MCA/FY BTECH)

SEMESTER : 7

PROGRAM: CSE-AIML

I / II / III / IV / V / VI / VII / VIII

DAY AND DATE: 08-10-2025 Wed

COURSE NAME: Deep Learning

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.



Space For Marks	Question No.	START WRITING HERE (Begin answer for each question on a new page)
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Q.1a

Soln.

$$f(w) = \frac{1}{2} \|Xw - y\|^2$$

i.e. $\eta < \frac{2}{\lambda_{\max}}$ $\lambda_{\max}$ is largest eigen value of $X^T X$.

gradient descent is used to minimize the loss in the dataset & update the values after every epoch.

Gradient descent for Stochastic g.d. is given by

$$d(\theta) = \frac{1}{f} (f(x_{t+1}, y_{t+1}))$$

$\eta \rightarrow$ learning rate is a hyperparameter that helps us to update input & bias value.



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Q2(a)

Soln. $\Rightarrow$ Convolutional neural n/ws are mostly used for image processing & shrinking the input image using kernels & filters. Pooling the most important technique here because it helps us to get generalized output. Various types of poolings are found such as max pooling, min pooling & average pooling. It is applied using aggregate functions on a specific window either we can pad the input or slide on it in a particular order. The size of the output depends on various factors such as padding, filtering, pooling filters etc. The formula to get the final output is given below.

$$O = \frac{n + 2p + k}{s}$$

Separable convolutions are critical in modern architectures because we want to compress our input & artificial neural n/w can't process thousands of inputs. So applying CNN filters before training is must since it provides the generalized input to us for evaluation. After that we can do whatever we want to our dataset & apply various learning techniques to get our output.


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 Question
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Q.1(a)

Soln. $\Rightarrow$ Overfitting occurs when model performs well on the training data but fails to perform on testing & validation data because it has learned fluctuations & noise while training due to which it fails to generalize when new input comes. The ability of model to generalize the input is declined. Overfitting can be reduced by generalizing the data, adding more features, scaling, one-hot encoding, min-max values etc.

Bias is high bias & low variance means model is overfitting.
 whereas low bias & high variance is case of underfitting.

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